

A Guided User Experience Using Subtle Gaze Direction

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Abstract—This paper demonstrates how illumination modulation can guide a user’s gaze on a display without them knowing. This technique takes advantage of eye biology and the differences in flicker fusion rate between the fovea and periphery, so that flickering in the periphery of a viewer’s attention attracts their gaze, yet is undetectable in their foveal vision. Applications for this technology include interactive media and gaming, visual training programs in medicine and defense, and advertising.

I. INTRODUCTION

In static images, human gaze is controlled by a variety of factors, including image contrast, presence of faces and other informative regions, and high edge density [1], [2], [3]. Some groups have altered images to attract a viewer’s gaze towards regions of interest by adding contrast or non-realistic features [4], [5]. However, digital screens that display dynamic images offer new possibilities. Previous research has shown that illumination modulation can direct a user’s gaze around a digital image [6]. This technique used active eye tracking and terminated the modulation before a user could focus on the modulating object.

However, by taking advantage of eye biology, a similar type of illumination modulation can be used to direct a user’s gaze without the modulation being noticeable in their foveal vision. Studies have shown that the flicker fusion threshold is different for foveal and peripheral vision [7]. Therefore, for images flashing at certain frequencies, the modulation will be apparent when the object is in the periphery, but the images will be fused together when viewed directly.

The applications of technology that can subtly direct gaze are varied. It would be valuable for visual training programs in medicine and defense, as experts show different eye patterns than novices. For example, a trained radiologist scans an MRI image differently than an apprentice doctor. Additionally, more interactive media and gaming experiences would be enabled. Overall, this research would augment knowledge of human perception.

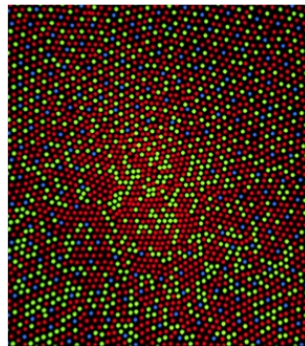


Figure 1. Distribution of red, green, and blue cones in the fovea. Courtesy of [8].

II. RELATED WORK

Eye biology and human visual perception are topics that have been extensively studied. The eye is comprised of two types of photoreceptors: rods and cones. Cones are concentrated at the center of gaze, and are responsible for color sensitivity. The three types of cones—red, green, and blue—are also asymmetrically distributed, with red and green cones being more numerous and blue cones being more sparse and located away from the center of gaze. Figure 1 shows how the different cone types are located across the eye. Rods are concentrated in the periphery of one’s gaze, and are extremely sensitive to illumination and motion. Figure 2 is a plot of the distribution of cone and rod cells as a function of the angle of gaze (relative to center).

In 2009, Bailey et al. introduced a novel technique to direct a viewer’s gaze about an image [6]. They used subtle modulation to attract the attention of a viewer, noting that peripheral vision responds to stimuli faster than foveal vision. By modulating regions of the scene in the periphery of a viewer’s vision, they caused the viewer’s eyes to saccade to that region. Luminance modulation and warm-cool modulation were chosen, as the human visual system is very sensitive to these changes [10]. A few groups have applied this method for medical training and visual searches [11], [12].

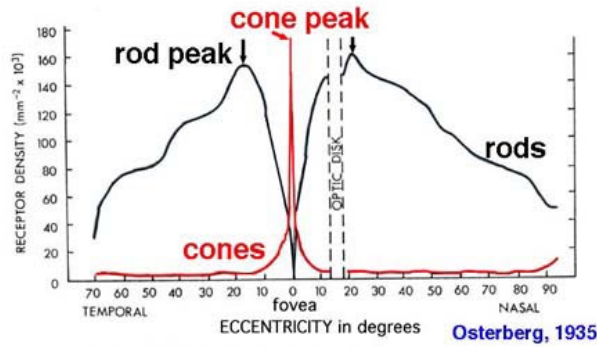


Figure 2. Distribution of rods and cones in the retina. Courtesy of [9].

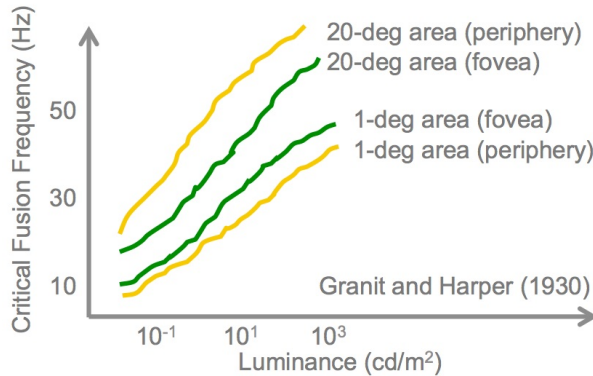


Figure 3. Critical fusion frequency vs. luminance. Adapted from Granit and Harper.

While this technique was successful, it used active eye tracking to detect when a viewer's eye was moving towards the modulating object, then stopping the modulation. This limits potential applications as it requires an eye tracker setup to be present. The method employed in our experiment takes advantage of differences in the flicker fusion threshold between the periphery and fovea, which was first investigated in the 1930's. Humans notice flickering images at low frequencies, but there is a threshold frequency at which flickering images fuse into one image. This critical fusion frequency depends in the luminance of the object (the Ferry-Porter law) and it's size (the Granit-Harper law). Granit and Harper demonstrated that the flicker fusion frequency depends on where the object is located [7]. For large objects, the critical fusion frequency is higher if the object is in the periphery of a viewer's gaze than if the object is centered in the viewer's fovea, as shown in Figure 3.

III. METHODS

A. Setup

In this paper, OpenGL was used to render images that were then displayed on a monitor that operates at

60 Hz and has a resolution of 1920x1080 pixels. The images shown were 1024x768 pixels. An eye tracker manufactured by The Eye Tribe was used to track the gaze of test subjects at 30 Hz. A chin rest was used to keep the viewers' heads steady at a fixed distance from the screen (approximately 2 feet). The experiments were all performed during the day but with the blinds closed, so the room brightness was kept roughly constant.

B. Subjects

Five male participants between the ages of 22 and 26 participated in this study. All participants had normal or corrected-to-normal vision and participated in the following tests.

C. Modulation Intensity, Frequency, and Color

Modulation intensity, frequency, and color are three parameters that drastically alter the effect of the gaze guiding. The teapots were chosen to be blue teapots, as people tend to have less blue cones than red or green cones, and blue cones are concentrated away from the gaze center.

According to the research on critical fusion frequency and our estimates of object size and luminance, frequencies around 30 Hz were likely to be optimal (above the flicker fusion threshold in the fovea but below it in the periphery). Given that the monitor was a 60 Hz monitor, 30 Hz was also the maximum frequency that could be displayed. Therefore, 30 Hz was chosen as the modulation frequency.

Modulation intensity is another critical parameter. If the modulation intensity is too high, it is noticeable in foveal vision, but if it is too small, then perhaps the effect is not apparent in the periphery. Additionally, modulation intensity changes the average luminance of the object, which in turn affects the flicker fusion frequency. The modulation intensity was 30% (high value/low value - 1) and this value was chosen based on our own perception. We tested different levels of modulation intensity with our subjects at the end of the experiment to see at what point flickering was noticeable in their foveal vision.

D. Gaze Direction Test

Figure 4 shows the image shown to subjects. First, the subjects were shown an image without illumination modulation as a control experiment. Their eye movements were tracked for approximately 10 seconds. Next, the subjects were shown our gaze-directing test. Again, the image showed six blue teapots, but this time, one teapot of the six was modulating: first the top left teapot, then the top middle teapot, and then the bottom right teapot. Figure 5 shows the intended path of a user's gaze during



Figure 4. Image of six identical teapots displayed to subjects.

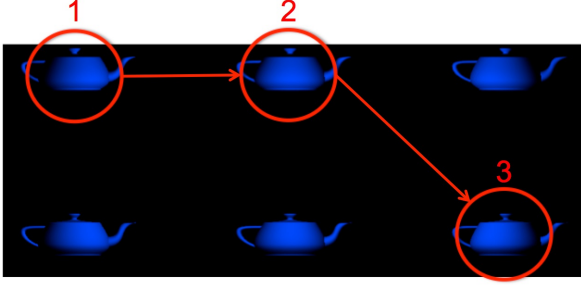


Figure 5. Intended path of a user's gaze during the test experiment.

the test experiment. The test was also for 10 seconds, with each of the three teapots modulating one third of the total time.

E. Analysis

The data from the eyetracker was parsed using a Python script and then analyzed using MATLAB. A viewer's gaze was calculated to be the average X and Y coordinates of the left eye and right eye. Points where the eye tracker was not able to compute an eye position (possibly due to the subjects blinking or looking too far away from the screen) were not included in the analysis.

One primary tool used to investigate the results was videos of the data. The videos contained significant information about how long subjects spent looking at each teapot, as well as their general eye patterns throughout the test.

Additionally, histograms were used to compare the test and control experiments. To compute the histogram, six bins were defined between the teapots as shown in Figure 6, and the number of points in each section were counted. Looking at the test vs. control ratio for each section demonstrates how the illumination modulation effects the gaze direction of the user.

IV. RESULTS

Eye-tracking data was gathered and analyzed for the five subjects. Figures 7 shows an example of the data from one of the subjects. In the control experiment, the subject mainly focused on the top right teapot and

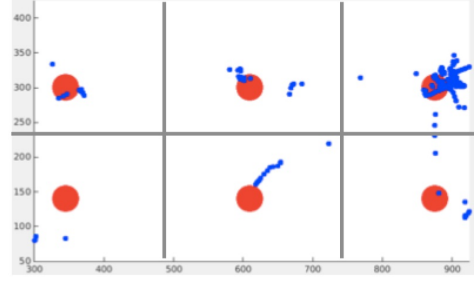


Figure 6. Example of how histograms are computed for the data.

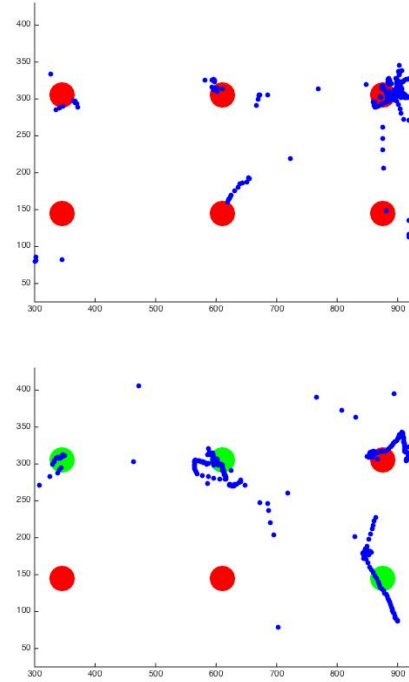


Figure 7. Eye tracking data from one subject. Control (top) and test (bottom).

spent a little time observing the other teapots. In the test experiment, the subject's gaze concentrated on the top left, top middle, and bottom right teapots, which is where his gaze was directed. Figure 8 shows how his gaze was directed from the top left teapot, to the top middle teapot, to the bottom right teapot.

Analysis on all of the test results (i.e. where the teapots had modulation) was first conducted, in which the number of data points in each of the six sections were counted. This initial heat map analysis showed that during the test, users tended to look at the sections where the teapots were modulating (sections 1, 2, and 6). Compiled across all five users, test-only analysis gave counts of 223, 470, and 342 for sections 1, 2, and

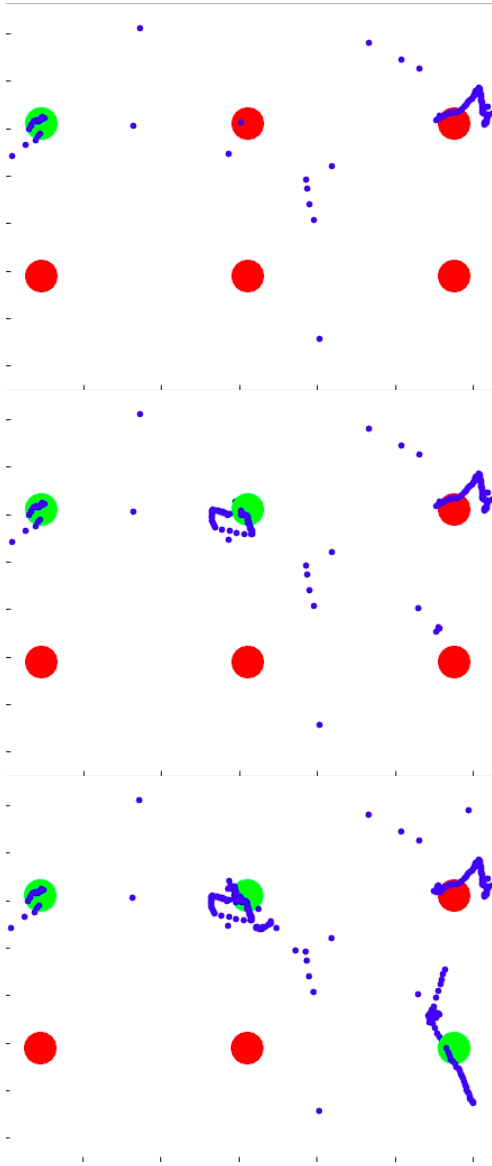


Figure 8. Eye tracking data for a subject. Test data after the top left teapot modulates (top), after the top middle teapot modulates (middle), and after the bottom teapot modulates (bottom). Note that the data accumulates over the three images.

6 respectively. The other three sections (3, 4, and 5) reported counts of 272, 79, and 190 respectively. This indicates that users tended to focus most on the top-middle teapot, which may have to do with its central location on the screen.

The second analysis performed computed ratios of test vs. control on the counts of data points per each of the six sections. Aggregated across all five users, sections 1, 2, and 6 show ratios of 2.28, 2.16 and 2.42, respectively, whereas sections 3, 4, and 5 show ratios of 0.85, 0.91, and 1.02, respectively. This indicates that users were at least two times more likely to look at the teapots



Figure 9. Heat map of test vs. control ratios

Modulation Difference	Initial User Detection
10%	0
20%	1
30%	2
40%	2
50%	-

Table I
MODULATION INTENSITY DIFFERENCE AT WHICH USERS NOTICED THE MODULATION IN THEIR FOVEAL VISION.

of interest compared to other teapots, when compared to control. Figure 9 shows the heatmap of the test vs. control ratios.

After running the experiment, users were asked if they noticed any differences between the test and control experiments. Most users responded that they did not notice any difference, or that they noticed slight flickering but that they were not sure about it. This is exactly what was intended: subtle gaze directing. Additionally, users were given a flicker sensitivity test, in which a single teapot had increasing differences in flicker modulation. The sensitivity test, shown in Table I, demonstrates that most users began to detect flickering (when looking directly at the image) when the difference in modulation intensity was 30% - 40%. This confirms that the intensity modulation chosen was fairly optimal.

V. CONCLUSION

Results from the experiment validated the initial hypothesis that flickering visible only in the periphery of a user's field of view can subtly guide their gaze to an area of interest. Test vs. control ratios indicate that users are over two times more likely to look at the desired teapots than other teapots without consciously noticing that they were doing so. Users subjectively responded that they did not notice any flickering, and the flicker sensitivity test validated that the modulation used during the experiment would not be directly visible in their foveal vision.

These results are a promising first step towards subtle gaze directing without the need for an active eye tracking system. We look forward to applying these techniques in more complicated images, such as art, web pages, and medical images.

VI. FUTURE WORK

Though this experiment was a successful proof of concept, it has opened up many additional avenues for improvement and additional research. First, a more accurate, high-frequency eye tracker should be used in future experiments. While the eye tracker used was suitable for the proof of concept, it was prone to be slightly inaccurate at times and required adjustments. These adjustments add in a layer of human error that should be removed.

Second, more experimentation should be done on colors, sizes, and modulation intensity. Testing how changes in illumination modulation influence guidance effectiveness is a logical step. Investigating how the size of the flickering object, specifically how small it can be and still induce the desired effect, would be another interesting study. If an image has multiple colors, can flickering only in the blue channel still guide a user's gaze, or do other colors work better? These are all questions that should be answered in future research.

Third, the applicability of the technology must be examined. Can a user's gaze be guided across a work of art or an MRI image? Investigating how this technique can create different experiences for user's looking at web-based media would be interesting. Additionally, we would like to investigate how illumination modulation affects the perceived quality of images. These avenues also need to be explored to better understand where this exciting technology can be applied.

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